

## Discussion 6

1. Suppose you are playing a dart game and the probability of hitting the bullseye on any given throw is 0.40 and the throws are independent.
  - (a) What is the probability that you would get three bullseyes in nine throws?
  - (b) What is the probability that you would get at least six bullseyes in nine throws?
2. Suppose a survey found that approximately 30% of students believe the quality of food in the campus dining hall is unsatisfactory. Let  $X$  represent the number of students in a random sample of  $n = 50$  who hold this belief. Assuming the survey results are still accurate, find the following probabilities using cdf tables.
  - (a)  $X$  is at most 25.
  - (b)  $X$  is equal to 14.
  - (c)  $X$  is at least 31.
3. A random variable  $X$  has a binomial distribution with mean 5 and variance 2.5. Find  $P(X=2)$ .
4. Sarah works as a sales representative selling house insurance over the phone. Only 5% of all calls result in a successful sale. Assume that the outcome of each call is independent of the others.
  - (a) If Sarah makes 10 calls, what is the probability that she makes exactly 3 sales?
  - (b) If Sarah makes 10 calls, what is the probability that she makes at least 2 sales?
  - (c) If Sarah makes 10 calls, what is the probability that she makes at most 2 sales?
5. Suppose we have two coins. Coin  $C_1$  comes up heads with probability 0.2 and coin  $C_2$  comes up heads with probability 0.7. We repeat this process 4 times:
  - Choose a coin with equal probability.
  - Flip that coin once.

Suppose  $X$  is the number of heads after 4 flips.

- a. What is  $E[X]$ ?
- b. What is  $\text{Var}(X)$ ?
- c. Based on the number of heads we get, we earn  $Y = \frac{2}{X+1}$ . What's  $E[Y]$ ?